

Yashraj Singh

Hyderabad, India | yashraj2509singh@gmail.com | 88813 84160 | linkedin.com/in/yashraj-singh | github.com/yashraj0888

Summary

Full-Stack Software Engineer with professional experience building AI-powered healthcare applications using React, Next.js, Node.js, FastAPI, PostgreSQL, TypeScript, and Python. Experienced in developing scalable web applications, real-time dashboards, REST APIs, and multi-agent AI systems that improve scientific and healthcare workflows. Passionate about building secure, high-performance software that delivers real-world impact.

Skills

Languages: Java, JavaScript, TypeScript, Python

Frontend: React.js, Next.js, HTML, CSS, Tailwind CSS

Backend: Node.js, Express.js, FastAPI, Spring Boot, REST APIs

Databases: PostgreSQL, MongoDB, MySQL

AI & LLMs: Google ADK, MCP Servers, Agentic AI Workflows, Prompt Engineering, Generative AI, LLM Applications

Tools: Git, Docker

Experience

Full Stack Software Engineer, Aganitha Cognitive Solutions – Hyderabad, India February 2025 – Present

- Built a **real-time monitoring dashboard** using **Next.js**, **Node.js**, and **PostgreSQL** for an AI chat platform, enabling centralized log aggregation, database tracing, and live interaction monitoring, significantly improving production debugging and incident resolution.
- Designed and developed scalable **FastAPI** backend services powering a **multi-agent AI research platform**, integrating Google ADK and MCP to automate hypothesis generation and scientific study orchestration across large biomedical datasets.
- Developed reusable backend APIs and dynamic frontend workflows supporting agent selection, intelligent forms, collaborative chat, and workflow orchestration, improving research productivity and overall platform usability.
- Integrated the AI platform with **OPIK** for real-time LLM tracing, evaluation, and observability, reducing model evaluation turnaround by **30%** and developer prompt testing effort by **80%**.
- Built a configurable web scraping and automated PDF generation platform using **React**, enabling persistent user preferences and reducing reporting effort by **40%**.
- Developed reusable **Next.js starter kits** and authentication libraries supporting multiple identity providers, accelerating internal project setup, improving authentication reliability, and promoting code reuse across engineering teams.
- Collaborated closely with AI researchers, scientists, and cross-functional engineering teams to deliver production-ready healthcare AI applications focused on performance, scalability, and developer experience.

Personal Projects

- **Mail Maps** :- Built an automated job search and recruiter outreach tool using Next.js, Node.js, TypeScript, PostgreSQL, Google Gemini AI, and Puppeteer to assist me for job search. The app scrapes live real time jobs, resolves protected job pages via headless Chrome, generates recruiter email contacts using AI pattern matching, and manages automated email campaign workflows. (Work in progress)
- **Real-Time Chat Application** :- Built and deployed a real-time chat application with authentication, media sharing, and low-latency Firebase messaging, achieving message delivery under 100 ms and increasing user engagement by 45% ([Click to Preview](#)).

Education

Shri Ramswaroop Memorial College of Engineering and Management, B.Tech in Computer Science Sept 2020 – May 2024

- CGPA: 7.06

Little Flower House, Class XII - 70.4% 2018

Glenhill School, Class X - 9.2 CGPA 2016